oxidiser transport, chemical kinetics, and heat loss (conductive, convective and radiative). The experiments done aboard the ISS dealing with this issue include the Flames Extinguishing Experiment (FLEX), Burning And Suppression of Solids (BASS) and the Cool Flames Investigation. How cool is that? Though the name is not that kinda cool, it's still pretty cool.

Starting from the bottom, the Cool Flames Investigation is still ongoing, based on a novel phenomenon observed in a previous experiment (FLEX). Scientists observed that some fuels initially burn very hot and then appear to go out, but even after radiative extinction

Outer space makes flames fat

they continue to burn at a much lower temperature. This wasn't predicted by theoretical models nor has it been observed before, and could be a detrimental hazard for fire emergencies in space. For earthlings, understanding these cool flames will give us better models and help us design higher efficiency fuels and engines. The BASS and FLEX experiments studied the combustion of various types and shapes of fuels in microgravity, under different conditions and air flow rates, etc. The data collected will help improve existing models based on observed spread rates. With adequate ventilation, fire can burn and spread just as dangerously in space as it does on Earth. Moreover, techniques for putting out flames need to take into account the geometry of the flame as well as characteristics of the extinguisher. Otherwise, many Earth-based extinguishing techniques can do more harm than good by feeding the fire rather than snuffing it out!

Honourable mentions

In 2011, NASA's YouTube channel announced a competition for high school students to submit ideas for experiments to be carried out on the ISS. That contest added a whole bunch to all the interesting tests aboard the ISS. Even without the input of imaginative youngsters, there are many more intriguing experiments lined up to be carried out on the space station. For the complete list, head to *http://dgit.in/NASAExpList* . We've already given you the top three, here's some of our other favourites:

1. **Gecko gripper:** Geckos have unique nanostructured protrusions on their feet that allow them to stick to any surface except PTFE. While the movies have already developed products mimicking this technology (Mission Impossible), students at Stanford University have already developed Gecko gloves. Now scientists are trying to do the same for outer space applications. Imagine robots crawling around conducting repairs on the surface of spaceships like R2-D2..
2. **3-D printing:** Here on Earth, 3D printers are nothing new, but it was not known whether the process would work in space. The only thing to do was try it out, and they did. The machine was sent to space and a prototype printed. The polymer had no trouble building up into the programmed shape. In the future, on long distance missions, any parts required for repair can be fabricated on demand using stock material.
3. **Crystal growth:** If you do visit the link provided above, you will see many experiments listed, related to various types of crystal growth. This is because the growth of a crystal from a seed is a delicate process. As it takes place molecule by molecule, the near absence of gravity can have a dramatic effect on the subsequent growth. In an experiment dealing with peptides, there was no significant difference in the type of patterns that were formed during crystallisation, however, the nanopatterns were found to be more sophisticated due to the reduced effects of convection and sedimentation in microgravity. Crytals grown in microgravity can be used as templates and can be brought back down to manufacture better semiconductor devices.
4. **Fluid dynamics:** It may be obvious that fluids behave differently in microgravity, but their behaviour as observed in that environment can be used to test theoretical models and provide intuitive education to children back here on our gravitationally rich home planet. To see what we are talking about, watch *http://dgit.in/NASAWaterExp* (all of Don Pettit's Youtube videos are worth watching since he's a great combination of scientist and funny guy!)
5. **Sociological studies:** Outer space is a unique environment. People on space missions have to live in strange environments, in close proximity to a few others and extremely far from everyone else. It's no understatement for astronauts on a mission, that their lives depend on each other. Everything else is different too, from daily routines and sleep cycles to their nutrition and exercise regimen. In fact, many academicians have begun to term the whole thing as 'space culture'. Other than develop a better and healthier (mostly physically) space culture, its study can also aid with other groups of people in similar situations such as prisons and old age homes.
6. **Latent viruses:** Scientists have found that the stress of spaceflight induces the activation of latent viruses in the bodies of astronauts. In particular, they have identified the Epstein-Barr (EBV) and Varicella Zoster (VBV) viruses, whose presence in the saliva samples of the astronauts increased after take-off.
7. **Twin studies:** While current spaceships don't fly fast enough to test the twins paradox, NASA has had the opportunity to observe the difference between monozygotic twins Mark and Scott Kelly, as Scott Kelly returned to Earth earlier this year, after spending 340 consecutive days in space. d